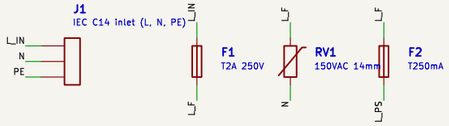
LINE SIDE: LIVE, 120 V

CONTROL SIDE: isolated, SELV (3.3 V / 5 V)

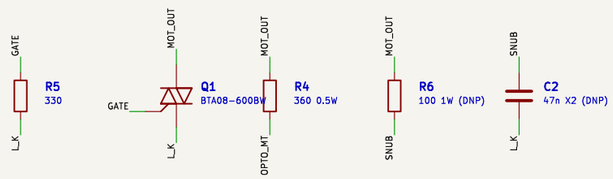
MAINS IN, FUSING, SURGE



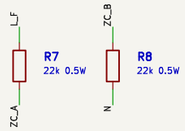
OUTLET (base's duplex: LIGHT always on, MOTOR controlled)



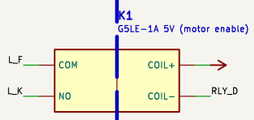
TRIAC STAGE (AN1662: triac in series with the motor)



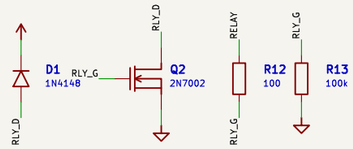
ZERO-CROSS DETECTOR (AN1662 uses a resistor to a pin; here isolated)



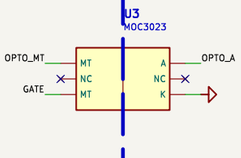
MOTOR ENABLE RELAY (cuts the triac's line on a fault)



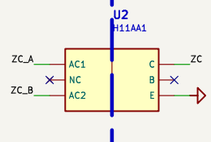
RELAY DRIVER



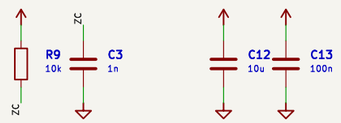
TRIAC FIRING (LED of U3)



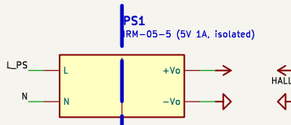
ZERO-CROSS INPUT



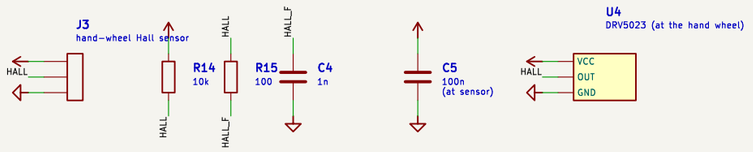
5 V BULK



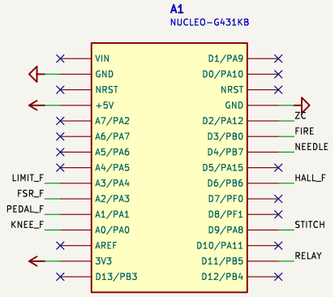
ISOLATED 5 V SUPPLY



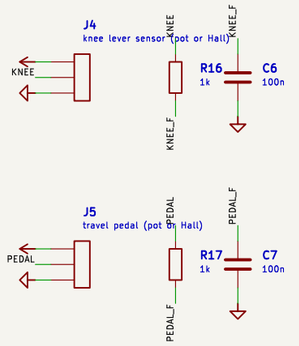
HAND-WHEEL HALL SENSOR (one magnet: one pulse per stitch)

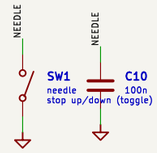


NUCLEO-32 (STM32G431KB)



OPERATOR INPUTS (RC-filtered ADC; switches: internal pull-ups)





NOTES (draft, Claude at Bob Drzyzgula's direction, October 2026; not built or tested)

1. After Freescale AN1662 (Skalka): zero-cross interrupt starts a timer; at the firing angle a short gate pulse (~100 us) fires the triac each half-cycle; tacho period measured by input capture; speed loop updated every half-cycle (120 Hz); stall watchdog. Added here: full isolation, relay, inputs.
2. Pins: ZC on D2/PA12 (EXTI); FIRE on D3/PB0 (TIM3_CH3, one-pulse); HALL on D6/PB6 (TIM4_CH1 input capture); RELAY on D11/PB5; A0–A3 = knee, pedal, FSR, speed limit; NEEDLE on D4/PB7 (SB2 must be OFF); STITCH on D9/PA8. D0/D1 left for the ST-LINK virtual COM port.
3. Power: PS1 feeds CN3-4 (+5V). Per UM2397 the ST-LINK is then unpowered: program and debug over USB with the mains unplugged. For live debugging use a USB isolator. Never connect a scope ground to the line side.
4. Fire only after a zero-cross has been seen and the speed command is zero (no start on power-up); close K1 last, open it first.
5. Values to check: R7/R8 set H11AA1 LED current (~3.9 mA peak at 170 V); R11 gives ~8 mA into U3 from 3.3 V; Q1 needs a heatsink for long runs; R6/C2 snubber is DNP with a snubberless (BW) triac and fitted if it false-fires; MOV 150 VAC for 120 V mains.
6. Hall: DRV5023 unipolar switch (open drain) at the hand wheel on a 3-wire cable; one magnet gives one pulse per stitch and needle position.

ISOLATION BARRIER: U2, U3, K1, PS1 straddle it; keep >= 6 mm creepage

Drafted by Claude at the direction of Robert Drzyzgula; not built or tested
After Freescale AN1662, with an isolated control side; NUCLEO-G431KB

Sheet: /
File: mcu_controller.kicad_sch

Title: VSM MCU phase controller (draft)

Size: A3	Date: 2026-10-22	**Rev: draft-1**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 1/1